City Resource and Infrastructure Management System (CRIMS)

1. Preface

This document specifies the software requirements for the City Resource and Infrastructure Management System (CRIMS).
It is intended for system users, developers, testers, and evaluators to understand what the system should do.

2. Introduction

2.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of CRIMS, based on stakeholder needs and system modeling using use-case and activity diagrams.

2.2 Scope

CRIMS supports city authorities in:

- Monitoring infrastructure
- Managing work orders
- Allocating budgets
- Scheduling and executing maintenance
- Tracking performance and reporting

Citizen complaint handling is treated as a separate subsystem under the same domain.

2.3 Definitions

Term	Description
CRIMS	City Resource and Infrastructure Management System
Work Order	Authorized request to perform maintenance
Stakeholder	User interacting with the system

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3. Overall Description

3.1 System Users

- City Administrator
- Department Officer
- Field Engineer

3.2 Product Perspective

CRIMS is a centralized management system that integrates:

- Monitoring
- Approval
- Scheduling
- Execution
- Reporting

It is not responsible for low-level hardware control.

4. User Requirements (High-Level)

UR1

The system shall allow the City Administrator to evaluate city project performance.

UR2

The system shall allow the City Administrator to approve work orders and allocate budgets.

UR3

The system shall allow the Department Officer to schedule inspections and track task progress.

UR4

The system shall allow the Field Engineer to inspect infrastructure and execute work orders.

UR5

The system shall allow the CFO to review budget allocations, review and accept vendor quotations, and approve procurement bills.

UR6

The system shall allow the QC Reviewer to independently review completed work and issue quality pass/fail certification.

UR7

The system shall allow Citizens and Public Representatives to submit infrastructure requests and track their status.

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5. Functional Requirements (System Requirements)

5.1 City Administrator Functions

- FR1: The system shall allow the administrator to monitor city resource usage.
- FR2: The system shall allow the administrator to approve work orders.
- FR3: The system shall allow the administrator to allocate budgets.
- FR4: The system shall generate performance reports.
- FR5 The system shall allow the administrator to give final budget approval after CFO review, validating against the annual department budget cap.
- FR6 The system shall allow the administrator to validate completed work before it proceeds to QC review.
- FR7 The system shall allow the administrator to forward budget allocations to officers for verification.

5.2 CFO Functions

- FR8 The system shall allow the CFO to review budget allocations after officer verification and approve, adjust, or reject them.
- FR9 The system shall require the CFO to provide an adjustment reason when the sanctioned amount is lower than the engineer estimate.
- FR10 The system shall allow the CFO to review vendor quotations before acceptance, ensuring government vendor preference is enforced.
- FR11 The system shall allow the CFO to approve procurement bills before payment, verifying GSTIN compliance.

5.3 Department Officer Functions

FR12: The system shall allow the officer to review assigned tasks.

FR13:The system shall allow the officer to report task progress.

FR14: The system shall allow the officer to schedule periodic maintenance inspections.

5.4 Field Engineer Functions

- FR15: The system shall allow engineers to inspect infrastructure.
- FR16: The system shall allow engineers to request required resources.
- FR17: The system shall allow engineers to identify and report new issues.
- FR18: The system shall allow engineers to execute approved work orders.
- FR19: The system shall allow engineers to perform maintenance activities.

5.5 QC Reviewer Functions

- FR20 The system shall allow QC reviewers to review completed work orders after administrator validation and issue pass/fail certification.
- FR21 The system shall require QC reviewers to provide comments when failing a review.
- FR22 The system shall enforce exactly one approved QC review per work order.
- FR23 The system shall require QC pass before a work order can be marked as completed.

5.6 Citizen / Public Representative Functions

- FR24 The system shall allow citizens to submit infrastructure requests with title, description, location (GPS), and priority.
- FR25 The system shall assign a unique public reference number to each citizen request for tracking.
- FR26 The system shall allow requests from multiple sources: citizen, public representative, sensor alert, and internal audit.
- FR27 The system shall notify citizens when their request is approved, converted to a work order, or rejected.

6. Non-Functional Requirements

6.1 Performance Requirements

- NFR1: The system shall respond to user actions within **3 seconds**.
- NFR2: Report generation shall complete within **5 seconds**.

6.2 Security Requirements

- NFR3: Only authorized users shall access system functions.
- NFR4: Role-based access control shall be enforced.

6.3 Reliability Requirements

- NFR5: The system shall ensure data consistency for work orders.
- NFR6: The system shall recover gracefully from system failures.

6.4 Usability Requirements

- NFR7: The system shall provide a simple and intuitive interface.
- NFR8: Users shall require minimal training to operate the system.

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7. System Models

This system is modelled using:

- **Use-Case Diagram** – to represent user goals
- **Activity Diagrams** – to represent workflow and decision logic
- **Sequence Diagram**

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8. System Evolution

This chapter describes the fundamental assumptions and anticipated changes to help system designers avoid restrictive decisions.

8.1 Fundamental Assumptions

- **Administrative Hierarchy**: The system assumes a three-tier administrative structure (Administrator, Officer, Engineer) for all workflow logic.
- **Connectivity**: It is assumed that Field Engineers have intermittent access to data networks while on-site.

8.2 Anticipated Changes

- **Hardware Evolution**: Integration with **IoT sensor networks** for automated infrastructure health monitoring.

- **Evolving User Needs**: Future requirements for **automated predictive maintenance** based on historical performance reports.

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9. Appendices

The appendices provide detailed, specific information regarding the hardware and data organization.

9.1 Hardware Requirements

Following the requirement for a centralized management system, the hardware must meet these configurations:

- **Server (Minimal)**: 8-core CPU, 32GB RAM, 1TB SSD for the centralized database.
- **Field Devices**: Mobile tablets with GPS capabilities and a minimum of 8 hours of battery life to support on-site inspections.

9.2 Database Requirements

This section defines the logical organization of data used by CRIMS:

- **Entity Relationships**: Each **Work Order** must be uniquely linked to a **Budget Allocation** and an assigned **Field Engineer**.

- **Persistence**: Historical data for all city locations must be preserved to allow for duplicate issue analysis.

10. Index

To ensure scannability for different stakeholders, the following indexes are included:

- **Index of Diagrams**:
    - Use-Case Diagram (Page 10)
    - Activity Diagrams (Page 11-17)
    - Sequence Diagram (Page 18-20)
- **Index of Functions**:

  City Administrator Functions (FR1–FR7)

  CFO(FR8-FR11)

  Department Officer Functions (FR12–FR14)

  Field Engineer Functions (FR15–FR19)

  QC Reviewer(FR20-FR23)

  Citizen(FR24-27)

11. Assumptions and Constraints

- The system operates within municipal administrative policies.
- Budget approvals follow predefined authority levels.
- Citizen complaint handling is modeled separately.

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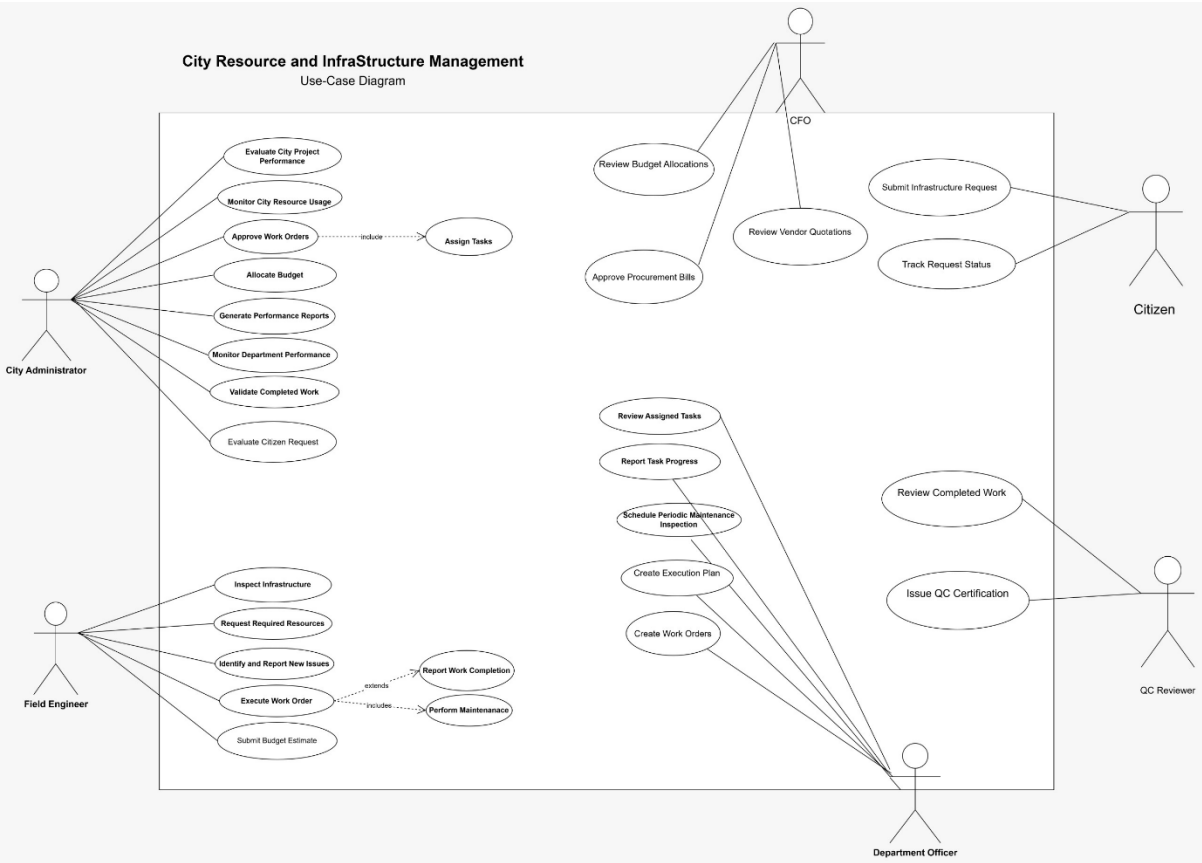
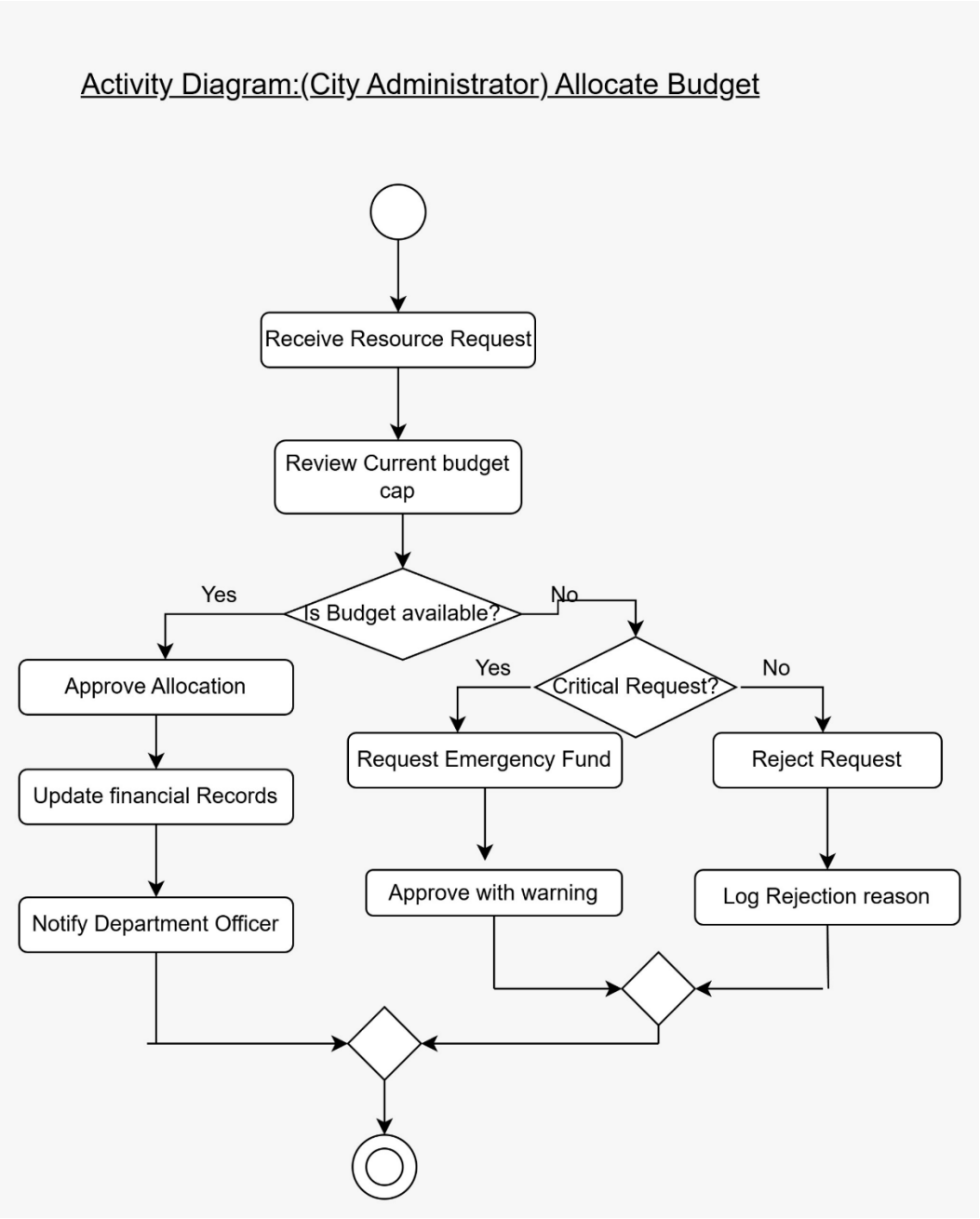
12. Conclusion

This SRS defines the functional scope and constraints of CRIMS, ensuring clarity, consistency, and traceability with UML models.

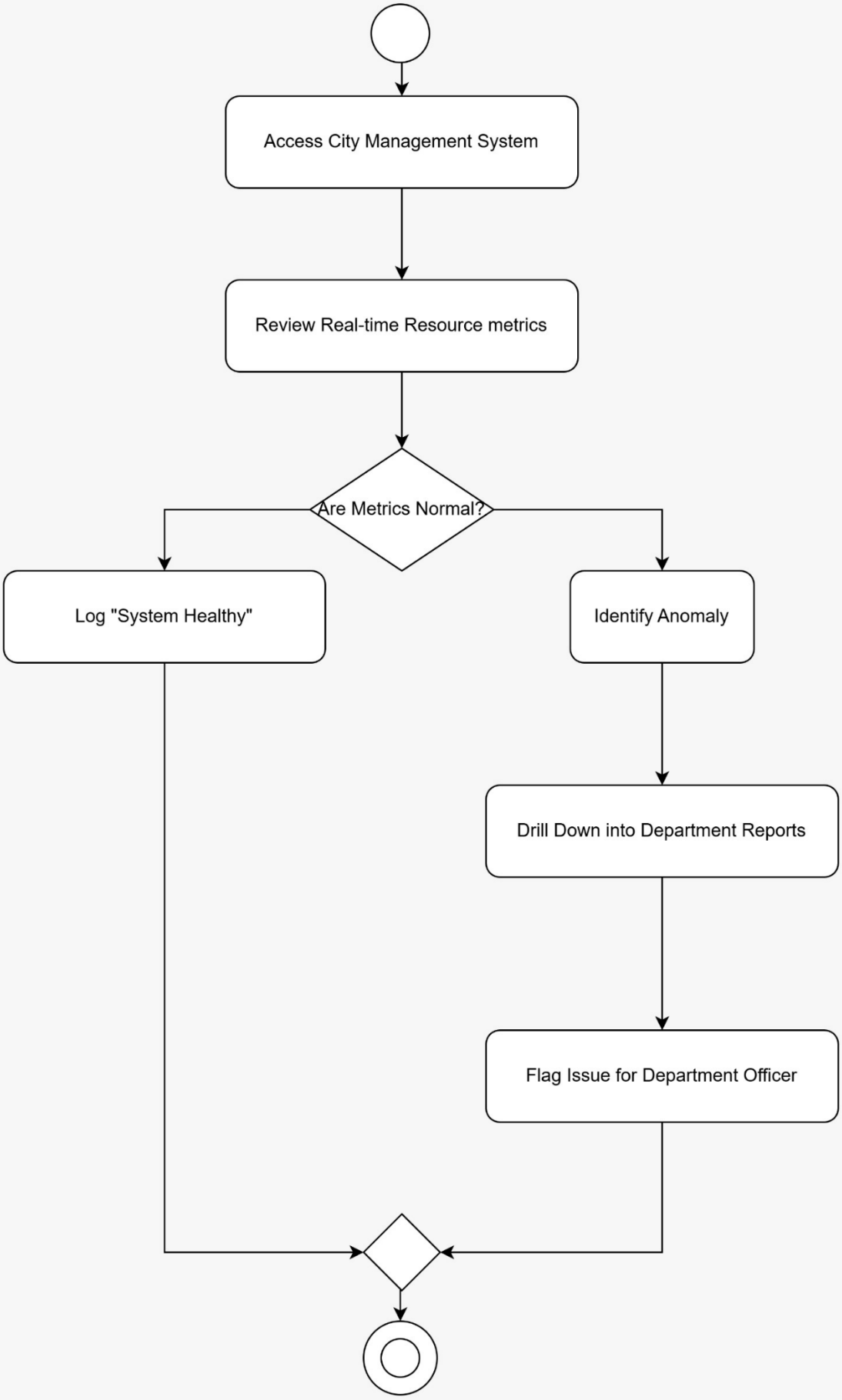
Use Case Diagram



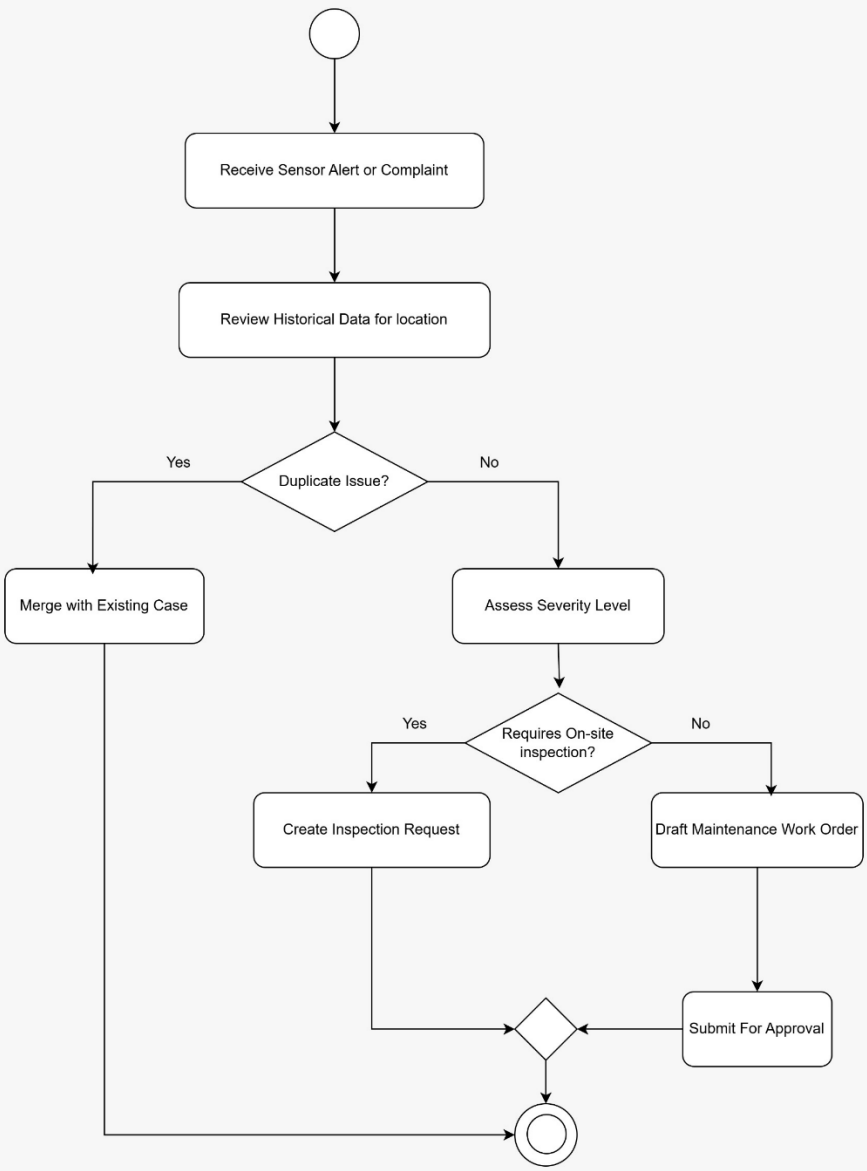
Activity Diagrams

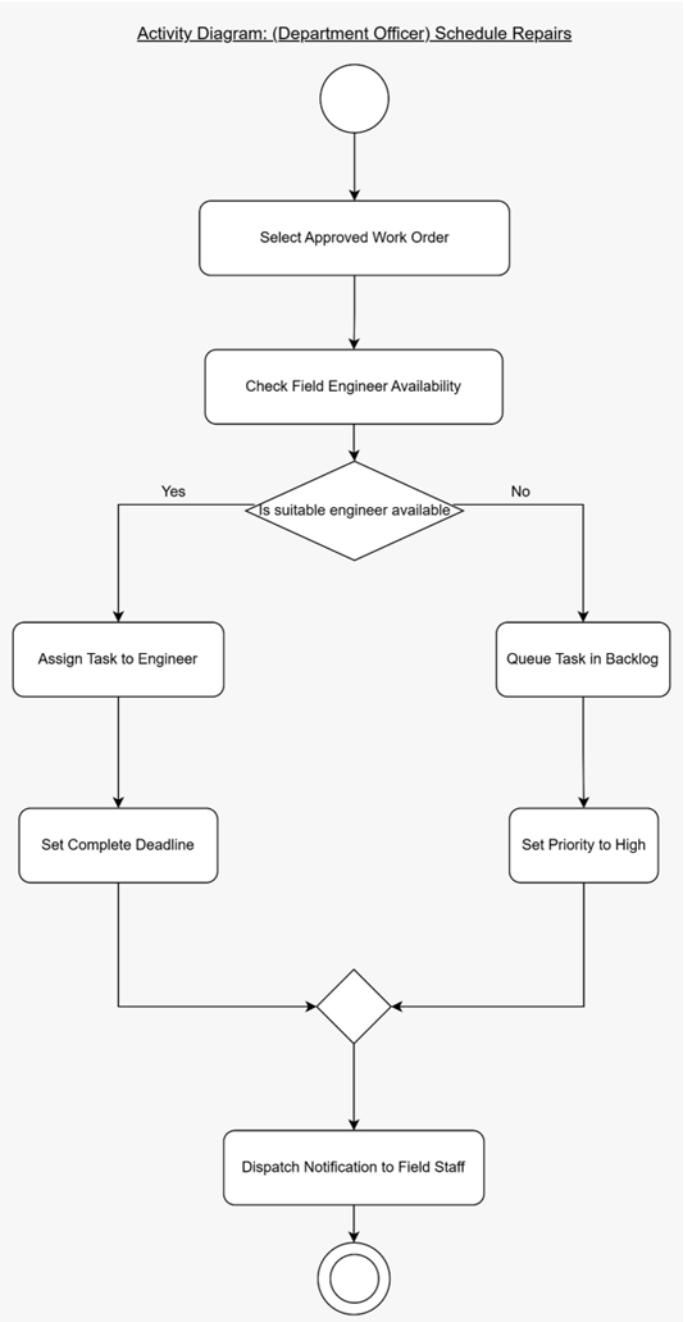


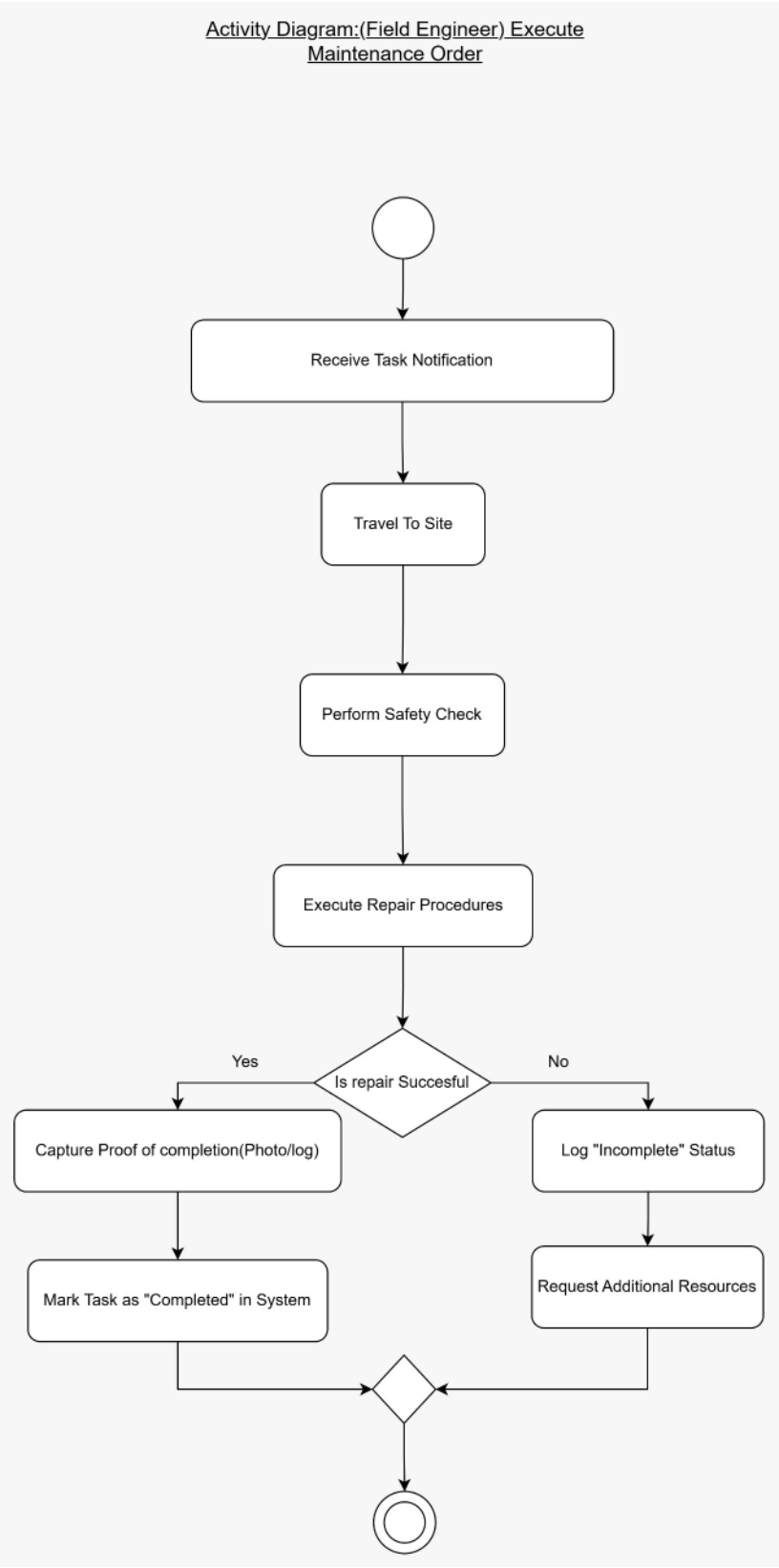
Activity Diagram:(City Administrator) Evaluate City Performance



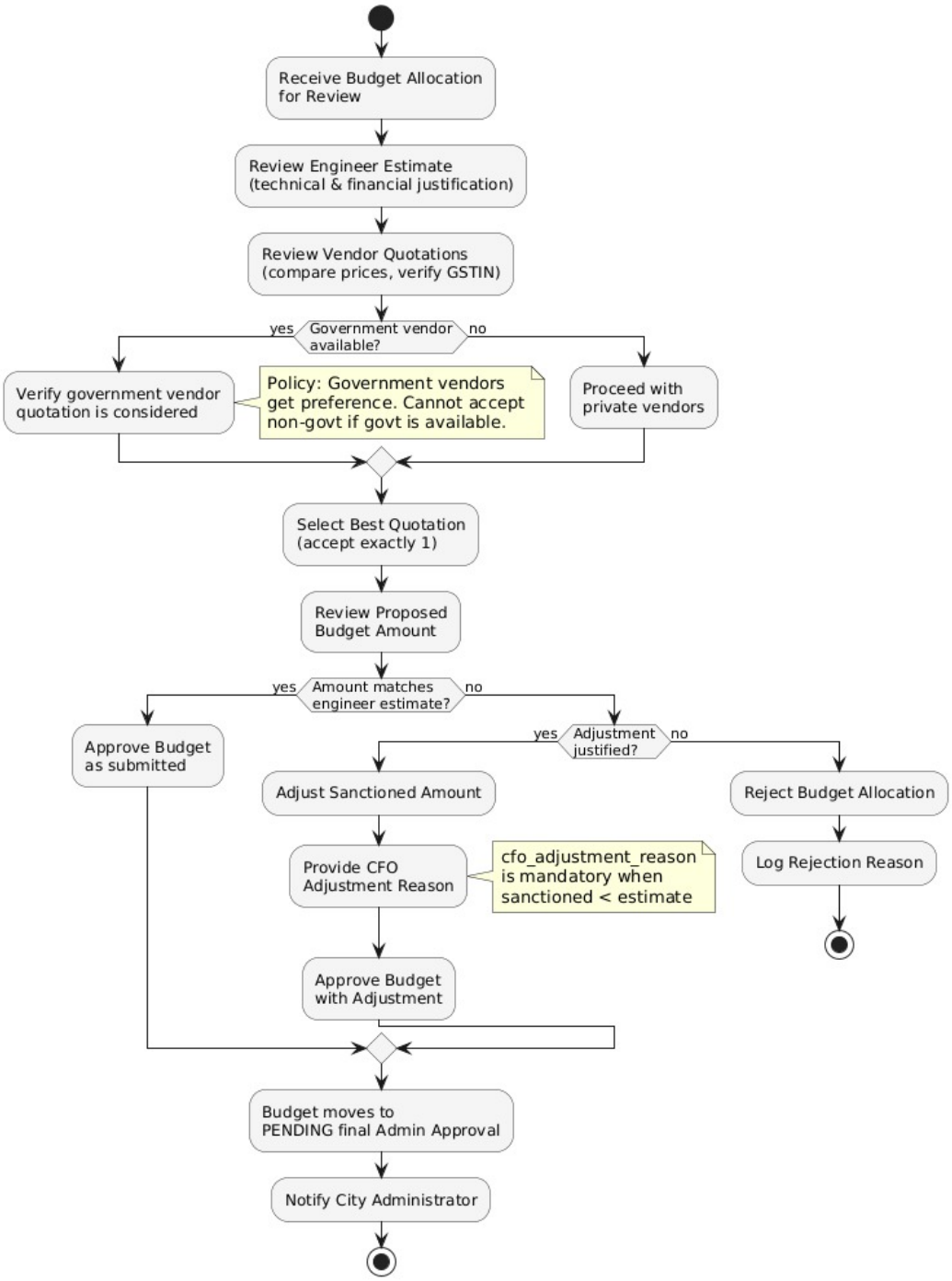
Activity Diagram:(Department Officer) Analyze Infrastructure Issue



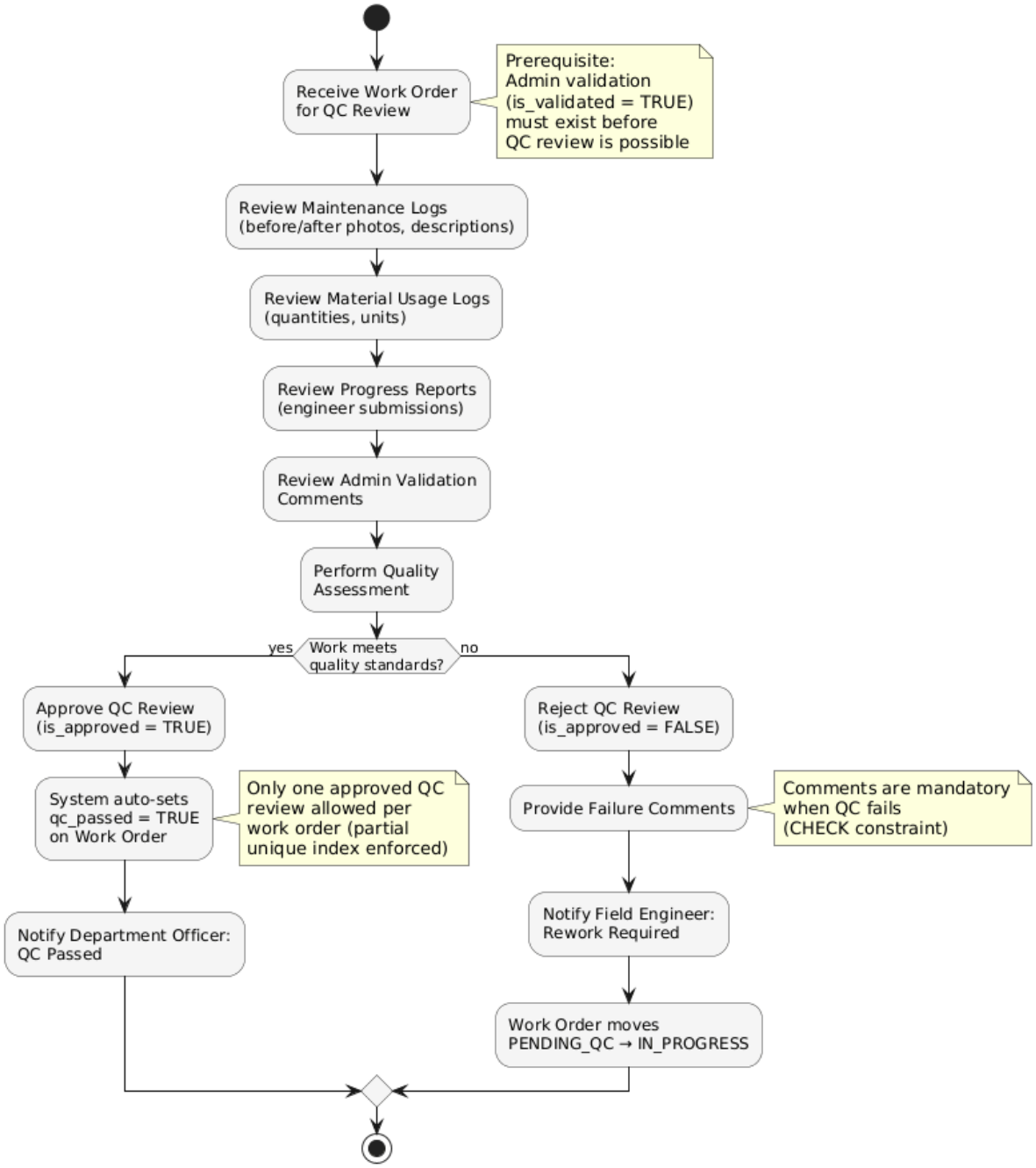

Activity Diagram: (Department Officer) Schedule Repairs
Select Approved Work Order
Check Field Engineer Availability
Is suitable engineer available
Yes
No
Assign Task to Engineer
Queue Task in Backlog
Set Complete Deadline
Set Priority to High
Dispatch Notification to Field Staff


Activity Diagram:(Field Engineer) Execute Maintenance Order
Receive Task Notification
Travel To Site
Perform Safety Check
Execute Repair Procedures
Is repair Succesful
Yes
No
Capture Proof of completion(Photo/log)
Log "Incomplete" Status
Mark Task as "Completed" in System
Request Additional Resources

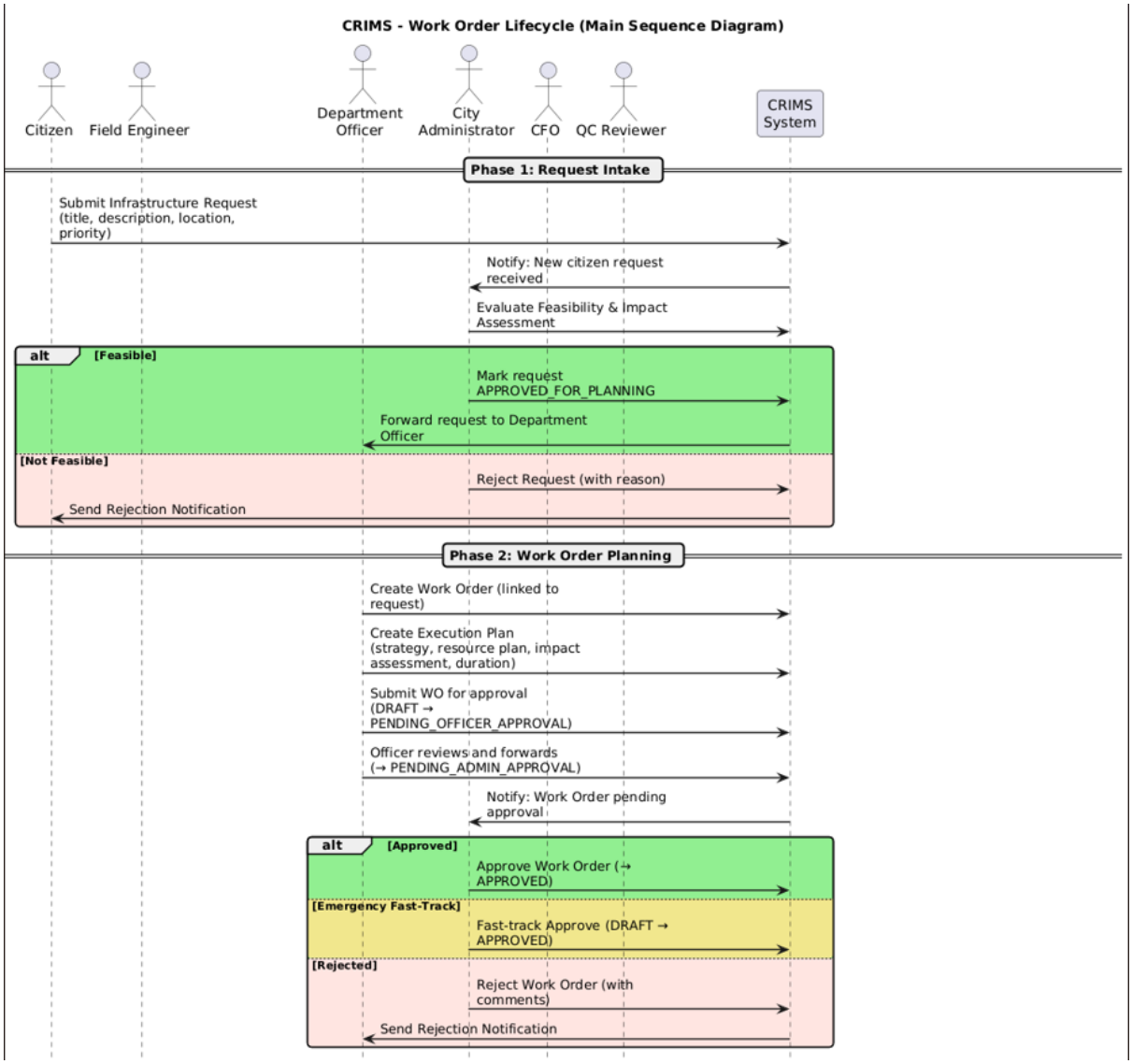
Activity Diagram: (CFO) Review Budget Allocation

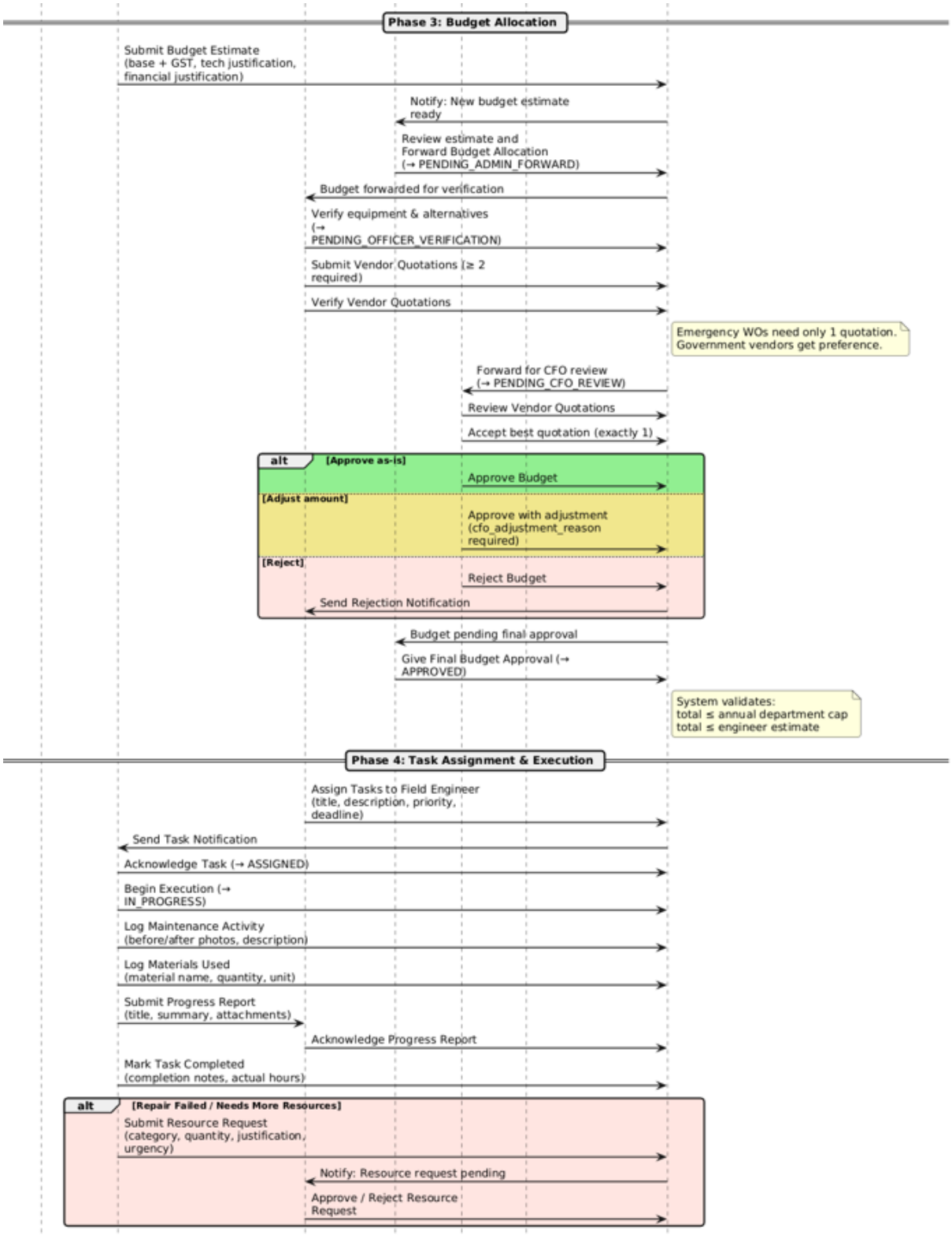


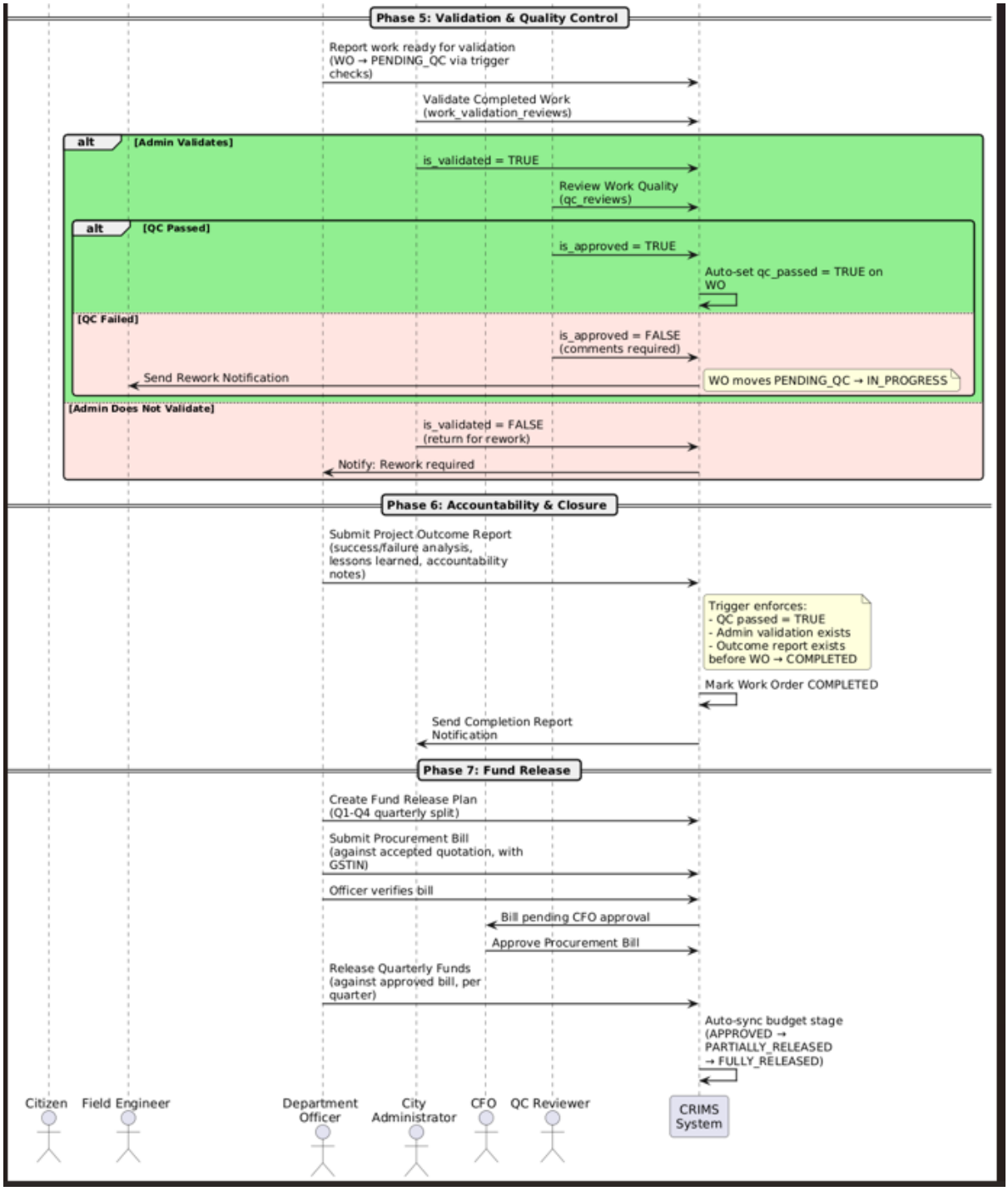
Activity Diagram: (QC Reviewer) Review Completed Work



Sequence Diagram




Phase 3: Budget Allocation
Submit Budget Estimate (base + GST, tech justification, financial justification)
Notify: New budget estimate ready
Review estimate and Forward Budget Allocation (→ PENDING_ADMIN_FORWARD)
Budget forwarded for verification
Verify equipment & alternatives (→ PENDING_OFFICER_VERIFICATION)
Submit Vendor Quotations (≥ 2 required)
Verify Vendor Quotations
Emergency WOs need only 1 quotation. Government vendors get preference.
Forward for CFO review (→ PENDING_CFO_REVIEW)
Review Vendor Quotations
Accept best quotation (exactly 1)
alt
[Approve as-is]
Approve Budget
[Adjust amount]
Approve with adjustment (cfo_adjustment_reason required)
[Reject]
Reject Budget
Send Rejection Notification
Budget pending final approval
Give Final Budget Approval (→ APPROVED)
System validates: total ≤ annual department cap total ≤ engineer estimate
Phase 4: Task Assignment & Execution
Assign Tasks to Field Engineer (title, description, priority, deadline)
Send Task Notification
Acknowledge Task (→ ASSIGNED)
Begin Execution (→ IN_PROGRESS)
Log Maintenance Activity (before/after photos, description)
Log Materials Used (material name, quantity, unit)
Submit Progress Report (title, summary, attachments)
Acknowledge Progress Report
Mark Task Completed (completion notes, actual hours)
alt
[Repair Failed / Needs More Resources]
Submit Resource Request (category, quantity, justification, urgency)
Notify: Resource request pending
Approve / Reject Resource Request


Phase 5: Validation & Quality Control
Report work ready for validation (WO → PENDING_QC via trigger checks)
Validate Completed Work (work_validation_reviews)
alt
[Admin Validates]
is_validated = TRUE
Review Work Quality (qc_reviews)
alt
[QC Passed]
is_approved = TRUE
Auto-set qc_passed = TRUE on WO
[QC Failed]
is_approved = FALSE (comments required)
Send Rework Notification
WO moves PENDING_QC → IN_PROGRESS
[Admin Does Not Validate]
is_validated = FALSE (return for rework)
Notify: Rework required
Phase 6: Accountability & Closure
Submit Project Outcome Report (success/failure analysis, lessons learned, accountability notes)
Trigger enforces:
- QC passed = TRUE
- Admin validation exists
- Outcome report exists
before WO → COMPLETED
Mark Work Order COMPLETED
Send Completion Report Notification
Phase 7: Fund Release
Create Fund Release Plan (Q1-Q4 quarterly split)
Submit Procurement Bill (against accepted quotation, with GSTIN)
Officer verifies bill
Bill pending CFO approval
Approve Procurement Bill
Release Quarterly Funds (against approved bill, per quarter)
Auto-sync budget stage (APPROVED → PARTIALLY_RELEASED → FULLY_RELEASED)
Citizen
Field Engineer
Department Officer
City Administrator
CFO
QC Reviewer
CRIMS System